

and $\rightarrow$ INTERSECTION ($\cap$)
 or $\rightarrow$ UNION ($\cup$)

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* TYPES OF EVENTS :

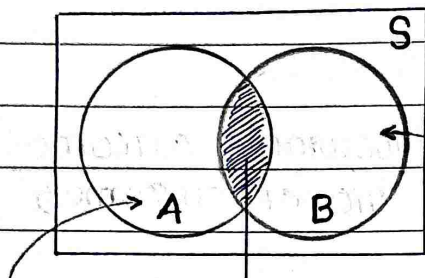
(a) JOINT EVENT

(b) DISJOINT EVENT

(c) DEPENDENT EVENT

(d) INDEPENDENT EVENT

(a) JOINT EVENT :- (non-disjoint / Non Mutually Exclusive Events)



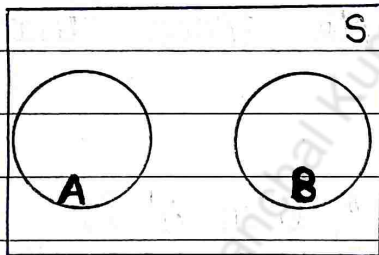
$\Rightarrow$ दो events के बीच सैम outcomes हों या दोनों events सैम time पर occur हो रहे हों।

Event A = common outcomes

$$\bullet P(A \text{ and } B) = P(A) * P(B)$$

$$\bullet P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$$

(b) DIS-JOINT EVENTS (Mutually Exclusive / Do not occur at same Time) :-



$$\bullet P(A \text{ and } B) = 0$$

$$\bullet P(A \text{ or } B) = P(A) + P(B)$$

* Example :- When a die is rolled we get events as follows :-

A = Getting even number = $\{2, 4, 6\}$

B = Getting prime number = $\{2, 3, 5\}$

2 common $\therefore$ JOINT EVENT

C = Getting ODD number = $\{1, 3, 5\}$

Comparing A and C, no common elements $\therefore$ DIS-JOINT EVENT.

(C) DEPENDENT EVENT :

One event depends on another or one event's action affects the other event.

Ex :- A = Drawing a red card

B = Drawing a black card

$$\therefore P(A) = \left(\frac{26}{52}\right)$$

$$\& P(B) = \left(\frac{26}{52}\right)$$

PROBABILITY OF B GIVEN A.

$$\therefore P(A \text{ and } B) = P(A) * P(B|A)$$

(d) INDEPENDENT EVENT :

A = Tossing a coin - {H, T}

B = Tossing 2nd coin - {H, T}

$$P(A \text{ and } B) = P(A) * P(B)$$

Q. Find the PROBABILITY of drawing a card that is red or a face card from a deck of a 52 cards.

Also name which type of event.

Soln 52 cards $\longrightarrow$ 26 red cards

$\longrightarrow$ 26 black cards

12 cards $\longrightarrow$ 6 red face card

$\longrightarrow$ 6 black face card

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$\therefore$ It's a Joint Event ; $\therefore P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$= \left(\frac{26}{52} + \frac{12}{52} - \frac{6}{52} \right)$$

$$= \frac{32}{52}$$

$$= \frac{8}{13}$$

Q2. Drawing a card that is a King or Queen.

Soln) $P(A \text{ or } B) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$: Disjoint Event.

CONDITIONAL PROBABILITY :-

→ PROBABILITY of an event based on the occurrence of previous event.

→ Previous event has already been occurred.

$P(A|B)$ = PROBABILITY of A, given B already occurred

$$\therefore P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

Example : MEDICAL DIAGNOSIS.

CONSIDER a medical scenario where we want to determine the probability that a patient has a specific disease (event A) given that they have tested positive for a certain medical test (event B).

→ The probability helps determine how likely a true test means a person actually has the disease. Think of it like testing whether an alarm means there's really a fire.

→ The example uses Baye's Theorem to calculate this.

- It's like asking, "Given a positive result, what's the chance the disease is really there?"
- We consider the overall disease rate, the accuracy of the TEST and the likelihood of testing positive.

* Baye's THEOREM helps us understand the probability of having a disease given a positive TEST RESULT.

Here's the breakdown :-

- $P(A)$ is the probability that a person has the disease.
- $P(B|A)$ is the probability of TESTING positive IF you have the disease.
- $P(B)$ is the probability of TESTING positive, WHETHER or NOT you have the disease.